

# Impact of Reward Function Selection on DQN-Based Traffic Signal Control

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**Abstract**—As the scale of traffic increases with growing urban populations, traditional fixed traffic signal timing methods become less effective in regulating traffic at intersections. Reinforcement learning (RL) methods can dynamically optimize traffic signal timing and lead to alleviated traffic conditions and have become a popular approach recently. This study aims to explore the impact of reward function selection in these RL approaches and the relationship between these selections. To conduct this study, three reward functions for a Deep Q-Network (DQN) were trained on a benchmark intersection: average waiting time, vehicle speed, and CO<sub>2</sub> emissions. From these simulations, we found that the DQN algorithm was able to reduce the mean waiting time by 82.63%, increase the mean speed by 146.84%, and decrease the mean CO<sub>2</sub> emissions by 4.26%, respectively, compared to a fixed-timing traffic signal controller. We also found that the RL-based model has a level of robustness over a traffic scale that deviates from training conditions. In addition to the impact of the reward function on the designated evaluation metrics, we also studied the impact on other metrics. We found that (1) the optimization of the average waiting time was linked to an increase in speed and a decrease in CO<sub>2</sub> emissions; (2) the optimization of the average speed was discovered to reduce waiting time and increase CO<sub>2</sub> emissions, and (3) the emission-targeted optimization produced an increased waiting time and decreased mean speed. These findings set expectations for DQN method performance for traffic signal control and indicate guidelines for reward function selection for multi-objective traffic control.

**Index Terms**—Traffic Control, Reinforcement Learning, Deep Q-Network, Reward Function, SUMO

## I. INTRODUCTION

Dense urban areas often suffer from high traffic congestion. Currently, about 55% of the global population lives in urban areas, and this is projected to rise to nearly 65% by 2050, making the issue even more significant [1]. This growth in traffic also leads to increased pollutants. In 2022, transportation accounted for 28% of greenhouse gas emissions in the US [2]. Regarding this challenging situation, autonomous vehicles could be a long-term solution to traffic congestion but only show significant improvement if over 39% of vehicles are autonomous [3]. Concerning emissions, zero-emission vehicles (ZEV) will serve as the permanent

solution [4]. This switch to ZEV will likely not be completed until 2050 [5]. Therefore, neither of these solutions will likely be fully implemented in the near future.

One promising approach to mitigating congestion and emissions in the interim involves optimizing traffic signals with existing infrastructures. The Los Angeles Department of Transportation demonstrated the impact of optimized traffic signals on the waiting time, speed, and CO<sub>2</sub> emissions of commuters through its study of the Automated Traffic Surveillance and Control System (ATSAC) on Los Angeles traffic. The department reported a 13% increase in travel speed, up to 43% decrease in travel delay, and up to 4% decrease in emissions as a result of the new system [6].

The study on alleviation of traffic congestion and reduction of emissions through traffic signal control has demonstrated promising results. By optimizing traffic signal timings, cities can improve traffic flow, reduce idle times, and lower overall emissions, contributing to a more efficient and sustainable urban environment. In [7], the authors proposed an ant colony algorithm for traffic signal timing optimization. In [8], a signal timing optimization method based on a genetic algorithm was proposed. In recent years, reinforcement learning (RL) methods have been widely applied to traffic signal timing optimization due to their capability to dynamically optimize traffic signal timing and lead to alleviated traffic conditions with highly dynamic and fluctuating traffic flows [9]–[11].

Reinforcement learning models “learn” with the aid of a reward function that evaluates the effectiveness of an action performed by the model. Depending on this reward, the neural network is updated via backpropagation. This process is continued until training is deemed to be completed. This is usually when the model is able to achieve a high reward consistently. Therefore, by properly configuring the reward function, reinforcement learning can be utilized to optimize specific behavior in a system [12]. Although a reward function may be employed with an emphasis on only one factor, this does not mean that others are not also affected. This is especially pertinent to the topic of traffic control [13] – as we expect all evaluation metrics to be improved at the same time. As evidenced by Los Angeles’s

This research was partially supported by the California State University Transportation Consortium Transport 2023 and 2024 programs.

ATSAC system, all three metrics – waiting time, speed, and emissions – were improved simply through the control of traffic signal timing. It is crucial to gain an understanding of the impact of these metrics and how they interact with each other, as it will allow for effective traffic control to achieve multiple objectives in the future. This study aims to explore the impact of different optimization objectives and provide insight into the connection between these objectives.

The main contributions of this work are listed as follows.

- We evaluated and compared the performance of fixed-timing signal and RL-based traffic control with respect to three evaluation metrics: average waiting time, vehicle speed, and CO<sub>2</sub> emissions.
- We analyzed the robustness of RL-based traffic signal control with respect to varying traffic scales.
- We studied the impact of the reward function in Deep Q-Network (DQN) on three evaluation metrics.

## II. METHODOLOGY

### A. Preliminaries

This section presents terminologies in reinforcement learning that are used in the remainder of this paper.

- **Neural Network:** A neural network can be thought of as layers of weighted nodes (called neurons) that are interconnected with weighted links [14]. The first layer is known as the input layer and is given data from the state of the environment. The last layer produces an output action that is implemented in the environment. When an RL agent “learns,” the weights of the nodes and links are updated via a process known as backpropagation to ensure that the desired results are achieved [14].
- **The Environment:** The environment is a representation of the conditions under which a reinforcement learning model (agent) is trained and tested. The environment typically provides some inputs (state information) that the model must handle to perform an action. The effect of the model’s action on the environment is what determines the reward that the model receives for use in backpropagation.
- **State Space:** The state space is a representation of all the possible conditions that the environment can be in. For any particular time, a specific state within the state space describes the current condition of the environment. This state is passed to the neural network for training or testing purposes [15].
- **Action Space:** The action space is the set of all actions that the model can take in response to the given state information [16].
- **Reward Functions:** Reward functions are utilized to “inform” the model how well it is performing. The behavior that the model optimizes is decided by the configuration of the reward function. The reward determines how the weights in the underlying neural network are altered in order to maximize performance [17].

### B. Optimization Objectives and Reward Functions

In this research, we aim to achieve three objectives through the optimization of traffic signal timing: 1) reduction in average waiting time, 2) increase in vehicle speed, and 3) reduction in the rate of CO<sub>2</sub> emissions. To achieve these objectives, we use a Deep Q-Network algorithm to optimize traffic signal timings in real-time. This approach allows for dynamic adjustments based on current traffic conditions, leading to more efficient traffic flow or lower emissions.

To implement this, data is collected from all vehicles that enter the simulation. Specifically, we use vehicles’ waiting time, speed, and CO<sub>2</sub> emissions rate to optimize traffic signal timings. The data is then fed into the DQN algorithm, which processes the information and determines the optimal signal timings to minimize waiting time, maximize vehicle speed, or minimize CO<sub>2</sub> emissions.

The following reward functions were chosen for the DQN algorithm.

- **Minimizing waiting time:** The reward function simply takes the negative value of the average waiting time of all the vehicles in the simulation. The equation for this can be written as  $r_t = -\Sigma W_t/N$ , i.e., the reward at time  $t$  is the sum of the waiting times of all vehicles present at time  $t$  divided by the total number of vehicles  $N$ .
- **Maximizing vehicle speed:** To maximize vehicle speed, the reward function is the average speed of all the vehicles present times a constant scaling factor, i.e.,  $r_t = (\Sigma S_t/N) * 100$ , where  $r_t$  is the reward at time  $t$ ,  $\Sigma S_t$  is the sum of the speeds of all the vehicles in the simulation, and  $N$  is the total number of vehicles.
- **Minimizing CO<sub>2</sub> emissions:** The negative value of the average rate of CO<sub>2</sub> emissions for each vehicle per timestep was taken as the reward for CO<sub>2</sub> emissions, i.e.,  $r_t = -(\Sigma E_t/N)/10$ , where  $\Sigma E_t/N$  is the average rate of CO<sub>2</sub> emissions for all vehicles in the simulation.

The state space in this work is denoted by the traffic conditions and the status of the traffic signal. The state was represented as an array of 21 values as defined by the default observation space in SUMO-RL [18], which is given by

$$\text{state} = [p, m, d, q] \quad (1)$$

where  $p$  represents the phase of the signal in the form of a one-hot vector,  $m$  is a binary value indicating whether the minimum amount of time per phase has elapsed,  $d$  is the lane density or the number of vehicles in a given lane divided by that lane’s capacity, and  $q$  is the lane queue or number of halted vehicles normalized with the lane capacity.

The action space of the model is the phases of the traffic signal, which are shown in Fig. 1. Here, all phases are encoded with a single integer ranging from 0 to 3. This gives a total of 4 actions that the traffic signal can perform to control the traffic signal.

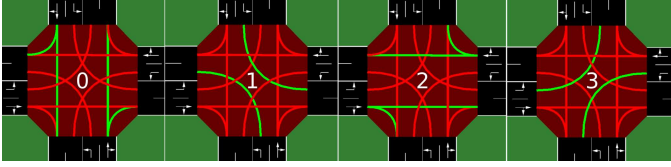


Fig. 1. Sketch of the intersection used in the simulation with all phases [18]. The diagram shows phases 0 to 3 from left to right.

### C. Evaluation of Results

1) *Benchmarks for Performance Comparison:* To obtain a benchmark for comparison, data was collected for a fixed-timing traffic signal with a cycle length of 100 s. This is within the range of 90-120 s cycle length that most traffic signals operate on in the Los Angeles area [6]. The average of this data over several runs was taken as the benchmark for mean waiting time, speed, and rate of CO<sub>2</sub> emissions.

2) *Evaluating Performance Under Various Traffic Scales:* Since the effectiveness of machine learning techniques is generally sensitive to the characteristics of the data [19], it is common to see reduced performance when testing the trained model. In the case of traffic signal control, the fluctuation in traffic flow can cause a significant difference in the performance of the algorithms. With this in mind, we test the robustness of the trained model against various traffic scales and examine the performance with respect to this variation.

3) *Impact of Reward Functions:* The selection of the reward function can significantly affect the performance of the different evaluation metrics. In this work, we selected the average waiting time, vehicle speed, and rate of CO<sub>2</sub> emissions as the reward functions and trained the models, respectively. In this process, we not only evaluate the metric corresponding to each reward function but also examine the other metrics to see how the optimization of one factor contributes to altered performance in others.

## III. CONFIGURATION OF SIMULATION

### A. Hardware

In the simulations, each model was trained and tested on an MSI GE76 Raider laptop with an NVIDIA GeForce RTX 3060 GPU (16GB RAM, 8-core CPU, 3840 GPU CUDA cores).

### B. Configuration of Simulation Environment

The environment employed in this work was a single intersection created in the traffic simulation software Simulation of Urban MObility (SUMO) [20]. The four-way intersection had a total of 8 lanes into and 8 lanes out of the junction. To model vehicles' emissions in SUMO, all vehicles were designated as passenger vehicles of emissions class HBEFA3/PC\_G\_EU4 [21]. This denotes a Euro norm 4-passenger gasoline car whose emissions are modeled using HBEFA3 guidelines [22].

The traffic flow was configured by defining the rate at which vehicles enter the simulation and the path they follow. For time from 0 to 25,000 seconds, vehicles enter from the north and south to go straight and right at a rate of 350 vehicles per hour. Left-turning vehicles from north and south enter at 300 vehicles per hour. The cross traffic from east and west is configured such that vehicles proceeding straight or right enter at a rate of 100 vehicles per hour, and vehicles taking a left have a rate of 50 vehicles per hour. At 25,000 seconds, the flow switches such that the vehicles from north and south adopt the traffic flow previously defined for east and west and vice versa.

### C. Training Process

In the process of training the traffic signal controller, we utilized the three reward functions defined in Section II-B. For each reward function, we trained the model until the average reward value didn't change significantly in new episodes. In each episode, 50,000 seconds in the simulation environment was evaluated. The selection of 50,000 seconds was due to the traffic in Los Angeles typically starting to increase at around 5 am and decreasing past 6 pm [23], which is a window of 13 hours or 46,800 seconds. This number was rounded up to 50,000 to simplify the plotting process.

### D. Testing Process

To determine the model's efficacy, the models were tested in the same environment with various traffic scales, where 1 represents an identical traffic flow rate to the one utilized in the training process, 0.25, 0.5, 0.75, 1.25, 1.50, 1.75, and 2 represents the corresponding flow rate compared to the training case. The scale can be regarded as the frequency that an additional vehicle will be added to the simulation environment.

## IV. RESULTS AND ANALYSIS

### A. Traffic Signal Control Using DQN Compared to Fixed-Timing

With the selected reward functions defined in Section II-B, we trained the traffic signal controllers via the DQN algorithm using each reward function individually. We compared the performance of the RL-based method employing different reward function selections with a benchmark test that uses a fixed-timing signal controller. It is worth noting that the scope of this work is to evaluate and compare how the selection of reward functions impacts the performance of the RL approach with the DQN algorithm from different evaluation metrics rather than proposing a new algorithm that competes with the existing methods.

Our results show that a converged reward value was obtained with all three reward functions. As shown in Fig. 2(a)(e)(i), the reward values quickly converged for all three reward functions. Beyond the convergence of the reward values, we can also see that the trained model achieved superior performance compared to the fixed-timing traffic

signal controller, where the average waiting time was reduced by 82.63% (benchmark: 21.82 s, DQN: 3.79 s), the average vehicle speed was increased by 146.84% (benchmark: 2.69 m/s, DQN: 6.64 m/s), and the mean rate of CO<sub>2</sub> emissions was reduced by 4.26% (benchmark: 2734.40 mg/s, DQN: 2617.91 mg/s), respectively.

### B. Evaluation of Trained Traffic Signal Controller

The traffic signal controller trained using DQN algorithms was tested with an identical traffic flow rate during the training process, and the results were compared with those of a fixed-timing signal controller. The average waiting time ranged from 3.7 seconds to 4.5 seconds, with the average being 4.09 seconds in our repeatability test. The average vehicle speed ranged from 6.2 m/s to 6.8 m/s, with an average of 6.56 m/s. The CO<sub>2</sub> emissions models had an average of 2619.96 mg/s, with the range for emissions being 2614 mg/s to 2628 mg/s. An example of the performance of the model that displayed results closest to the average can be seen in Figure 3. The trained controller was also able to handle the switch at 25,000 seconds more effectively than the fixed-timing method, as evidenced by the comparatively smaller change in performance at this point compared to the benchmark.

### C. Evaluation on Varying Traffic Scale

To determine the robustness of the trained traffic signal controller, we tested it with different traffic scales. As shown in Fig. 4, the DQN controller trained for average waiting time outperformed the fixed-timing controller when the traffic scale is between 75% and 200% of the baseline traffic flow (the one used in the training process). The trained controller also outperformed the fixed-timing controller in terms of the average speed of vehicles and CO<sub>2</sub> emissions with almost all traffic scales. This shows that the trained model is capable of handling dynamic traffic flow scaling even if such situations are not considered during the training process.

### D. Impact of Reward Functions

We also observed how the selection of the reward function impacts all evaluation metrics from Fig. 2.

1) *Waiting Time*: The first observation is that the decrease in average waiting time correlates with increases in mean speed through the junction when the reward function is designed to optimize average waiting time. However, the increase in speed is less pronounced than when the reward function prioritizes average speed. As shown in Fig. 2(b) and 2(e), the trained controller using the average waiting time as the reward function was able to reach a max speed of around 3.6 m/s, while the max speed using the average speed as the reward function was greater than 6 m/s. This result is intuitive in the sense that if vehicles spend less time waiting at a signal, they would also be able to maintain a higher average speed.

Regarding the performance in terms of reduction of CO<sub>2</sub> emissions, the average rate of CO<sub>2</sub> emissions is also below the baseline (Fig. 2(c)), which indicates that the optimization of average waiting time can give rise to a reduction in CO<sub>2</sub> emissions as well. Since the emissions are calculated through a combination of the speed and acceleration of the vehicle [22], the value of the CO<sub>2</sub> emissions is more scattered rather than fully converging to a steady value.

2) *Speed*: When the average speed is selected as the reward function, the average waiting time is also significantly reduced (Fig. 2(d)). Although this reduction is less than the performance achieved when the reward function is for waiting time (Fig. 2(a)), it is still a remarkable improvement in terms of the waiting time. However, the corresponding values of CO<sub>2</sub> emissions are clearly above the benchmark (Fig. 2(f)). This can be taken as proof that mean speed optimizations have a direct negative impact on CO<sub>2</sub> emissions.

3) *CO<sub>2</sub> Emissions*: When the reward function places emphasis on CO<sub>2</sub> emissions, a longer average waiting time and slower mean speed were observed (Fig. 2(g) and Fig. 2(h)). This could be due to the complexity of the emissions model, which involves multiple factors such as vehicle speed and acceleration, and the nonlinear relationship between these factors [22], [24]. As a result, various combinations of states could lead to a local minimum value of the reward function rather than the desirable one, resulting in a signal controller that reduces emissions but does not reduce average waiting time or increase vehicle speed.

## V. CONCLUSION AND DISCUSSION

Traffic control is of prime importance as urban traffic increases with population growth [1]. Consequently, commuters are spending more time waiting at traffic signals, with most urban commuters spending over 55 hours, or more than two days, in traffic annually [25]. To address this growing problem, intelligent adaptive traffic signals could offer a viable solution. By leveraging reinforcement learning techniques, we can develop traffic signal controllers that meet the specified requirements for effective deployment.

This study utilized the DQN algorithm to train traffic signal controllers using three different reward functions and tested the controllers in the SUMO environment to better understand the efficacy of such a system. This research found the average waiting time was decreased by 82.63%, the average vehicle speed increased by 146.84%, and the average CO<sub>2</sub> emissions decreased by 4.26% for a single 4-way intersection using the DQN algorithm, respectively, compared to a fixed-timing traffic signal controller. The trained controller also shows a certain level of robustness with respect to a range of traffic scales. This indicates that the trained controller is capable of handling dynamic traffic flow scaling even if such situations were not considered during the training process.

In addition to the impact of the reward function on the designated evaluation metrics, we also examined its effect on other metrics. We found that (1) optimizing average

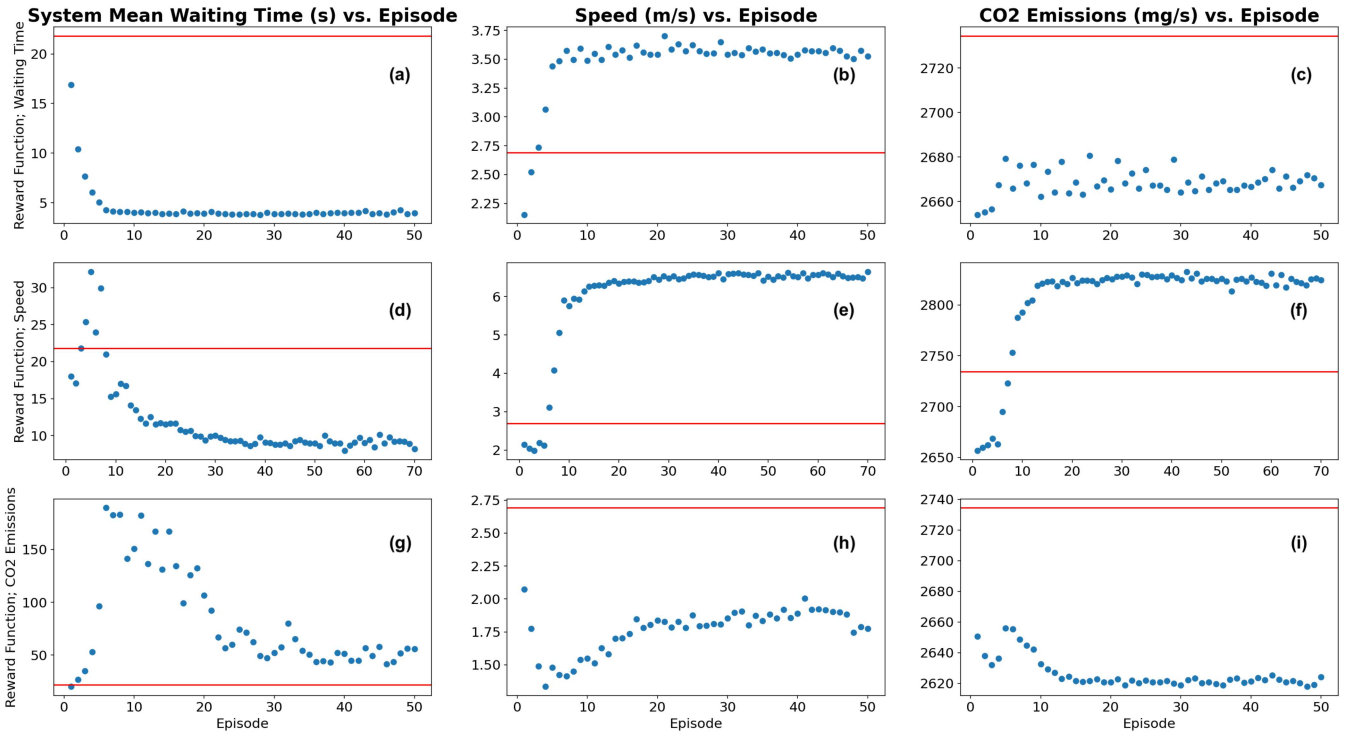


Fig. 2. Plots of training data. Columns represent specified factors vs. episodes. Rows represent plots for specified reward functions. The blue dots represent the data for each episode of the training process. The red line on each graph represents the benchmark performance using a fixed-timing signal controller.

waiting time (average waiting time is the reward function) led to an increase in vehicle speed and a decrease in CO<sub>2</sub> emissions, (2) optimizing average speed (average vehicle speed is the reward function) resulted in reduced waiting time but increased CO<sub>2</sub> emissions, and (3) targeting emission reduction (average emission is the reward function) caused an increase in waiting time and a decrease in mean speed. This result provided us with a fundamental understanding of how to select reward functions when multiple objectives are expected.

This study focused on the effect of traffic signal control on a single intersection. The rewards functions utilized provided rewards based only on one of the three factors discussed: waiting time, speed, and CO<sub>2</sub> emissions. For future research, a system of multiple intersections with signals controlled by a DQN algorithm could be explored. In addition, reward functions of factors alongside multi-objective reward functions that can effectively manage more than one target [26] could be studied. Further development in these areas will allow for a better understanding of the capabilities and limitations of RL methods. This, in turn, would bring such methods closer to being implemented in a tangible manner.

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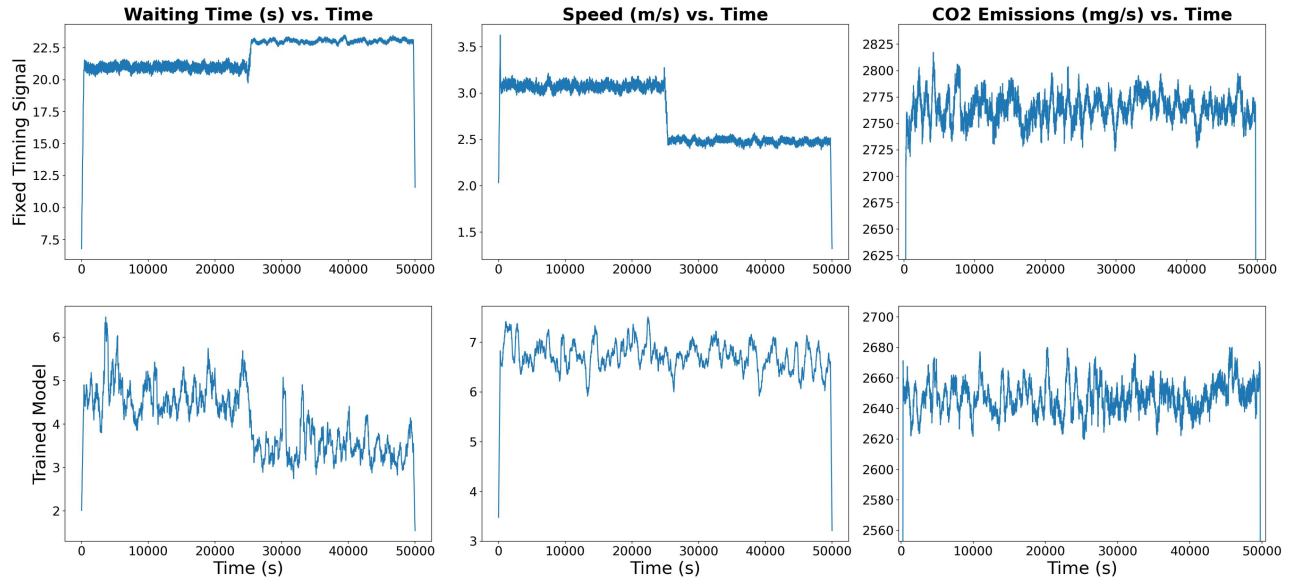


Fig. 3. An example of the fixed-timing method and DQN trained model being evaluated (closest to average) over 50,000-time steps. Data is smoothed over 100 steps. Column 1: average waiting time as reward function trained controller compared to the fixed-timing; Column 2: average speed as reward function trained controller compared to the fixed-timing; Column 3: CO<sub>2</sub> emission as reward function trained controller compared to the fixed-timing.

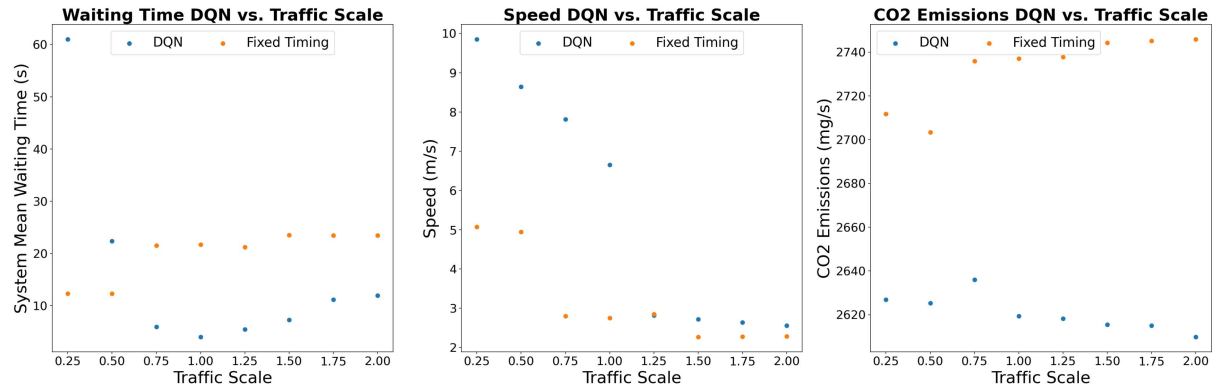


Fig. 4. Comparison between DQN trained signal controllers and fixed-timing configuration with respect to different traffic flow scales: (a) controller trained with the average waiting time as the reward function, (b) controller trained with the average speed as the reward function, and (c) controller trained with CO<sub>2</sub> emission as the reward function.

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